Dataset Specification & Clinical Attributes

Diabetes 130-US Hospitals (1999–2008) Dataset Documentation

Source Location: data/diabetic_data.csv & data/IDS_mapping.csv

Dataset Name	Diabetes 130-US Hospitals for Years 1999–2008
Source & Repository	UCI Machine Learning Repository (Donated by Strack et al., 2014)
Repository URL	https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008
Data Format	Tabular CSV (101,766 encounter records, 50 features)
Patient Cohort	71,520 unique diabetic patients admitted to 130 United States hospitals
Encounter Scope	Inpatient hospital encounters lasting between 1 and 14 days
Data Storage	Normalized into SQLite/PostgreSQL 3NF schema (healthcare.db)

Key Data Fields & Clinical Semantics

Feature Name	Data Type	Description & Clinical Significance
encounter_id	Integer	Unique identifier for hospital admission encounter (101,766 total).
patient_nbr	Integer	Unique identifier linking repeated encounters to the same patient (71,520 unique).
race, gender, age	Categorical	Patient demographics (Age bracketed in 10-year intervals).
admission_type_id	Integer (FK)	Mapped to Emergency, Urgent, Elective, Newborn, Trauma Center.
discharge_disposition_id	Integer (FK)	Mapped to Discharged to home, SNF, Expired/Mortality, Hospice.
time_in_hospital	Integer	Length of Stay (LOS) in days, ranging from 1 to 14 days (Average: 4.40).
num_medications	Integer	Total distinct medications administered during encounter.
num_lab_procedures	Integer	Total laboratory tests ordered during hospital stay.
diag_1, diag_2, diag_3	ICD-9 Codes	Primary, secondary, tertiary diagnoses grouped into 9 clinical classes.
readmitted	Categorical	Outcome target: '<30' (11.16%), '>30' (34.93%), or 'NO' (53.91%).

Data Privacy & HIPAA Considerations

In accordance with HIPAA guidelines and UCI data governance, all direct personal identifiers (e.g., patient names, social security numbers, street addresses) were anonymized prior to public donation. Patient ages are bracketed into 10-year intervals, and encounters represent historical aggregate hospital episodes. The system treats all data as educational Protected Health Information (PHI), implementing strict role-based authorization and audit logging.